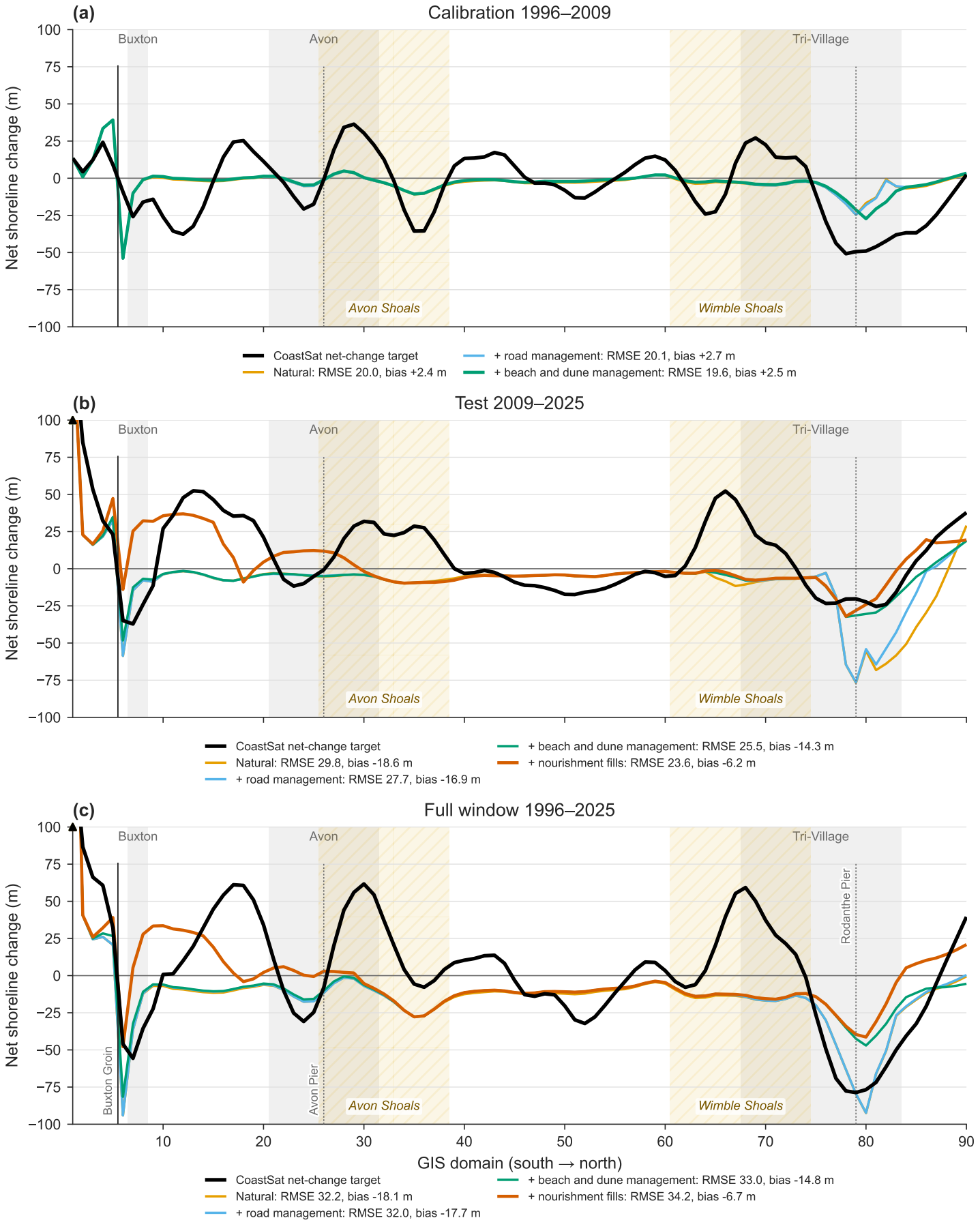

(a)
Calibration 1996–2009
Buxton
Avon
Tri-Village
Avon Shoals
Wimble Shoals
Net shoreline change (m)
CoastSat net-change target
Natural: RMSE 20.0, bias +2.4 m
+ road management: RMSE 20.1, bias +2.7 m
+ beach and dune management: RMSE 19.6, bias +2.5 m
(b)
Test 2009–2025
Buxton
Avon
Tri-Village
Avon Shoals
Wimble Shoals
Net shoreline change (m)
CoastSat net-change target
Natural: RMSE 29.8, bias -18.6 m
+ road management: RMSE 27.7, bias -16.9 m
+ beach and dune management: RMSE 25.5, bias -14.3 m
+ nourishment fills: RMSE 23.6, bias -6.2 m
(c)
Full window 1996–2025
Buxton
Avon
Tri-Village
Buxton Groin
Avon Pier
Rodanthe Pier
Avon Shoals
Wimble Shoals
Net shoreline change (m)
GIS domain (south → north)
CoastSat net-change target
Natural: RMSE 32.2, bias -18.1 m
+ road management: RMSE 32.0, bias -17.7 m
+ beach and dune management: RMSE 33.0, bias -14.8 m
+ nourishment fills: RMSE 34.2, bias -6.7 m